

SYSTEMS SIMULATION

Systems Analysis & Design

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UNIVERSIDAD DISTRITAL
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Outline

- 1 Basic Concepts
- 2 Cellular Automata
- 3 Digital Twins



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What is a simulation?

- **Simulations** are a type of real-world representation, **but rarely exactly** like the real world.
- Sometimes you need to **test** or **experiment** with **expensive use cases**. Simulations let you **experiment** with different inputs, conditions, and hyperparameter optimizations.
- Also, there are dangerous or hard-to-reach **scenarios** where **simulations** become the best option.
- You should define the **level of detail** well enough to represent the expected behavior, without failing due to either **high complexity** or **oversimplification**.



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Events and Stochastic Processes

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Case Study: Load Balancer in a Cloud Architecture

- A **load balancer** distributes **incoming requests** across multiple **server instances** to optimize performance and availability.
- **Simulation** helps predict **system behavior** under different traffic loads and failure scenarios.
- Key metrics include **response time**, **throughput**, and **resource utilization**.



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Cellular Automata

- **Cellular Automata** are a **discrete model** consisting of a **grid** of cells, each with a **state**.
- The **state** of each cell is updated based on the **states of its neighbors**.

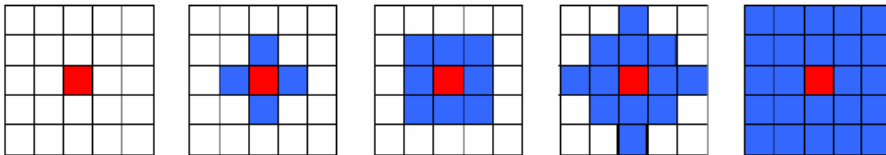


Figure: Cellular Automata Typical Neighborhoods



Game of Life

- **Conway's Game of Life** is a **cellular automaton** devised by the British mathematician **John Horton Conway** in 1970.
- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.
- Rules:
 - Any live cell with fewer than two live neighbors dies (underpopulation).
 - Any live cell with two or three live neighbors survives to the next generation.
 - Any live cell with more than three live neighbors dies (overpopulation).
 - Any dead cell with exactly three live neighbors becomes alive (reproduction).



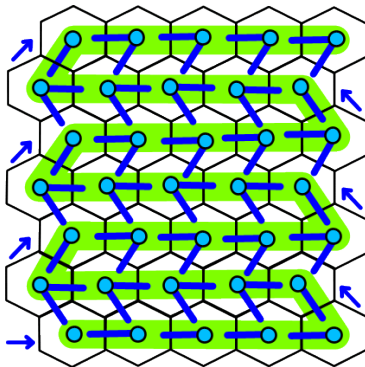
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- It is a **zero-player game**, meaning that its **evolution** is determined by its **initial state**, requiring no further input.
- Rules:
 - Any **live** cell with **fewer than two live** neighbors **dies** (underpopulation).
 - Any **live** cell with **two or three live** neighbors **survives** to the next generation.
 - Any **live** cell with **more than three live** neighbors **dies** (overpopulation).
 - Any **dead** cell with **exactly three live** neighbors becomes **alive** (reproduction).



Honeycomb Cellular Automata

- **Honeycomb Cellular Automata** use a different topology where each cell has **six neighbors**.
- This representation has different **dispersion properties**, which can be more interesting for certain applications.



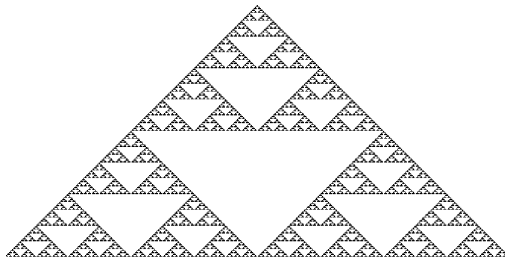
SIR Model

- The **SIR Model** is a **compartmental model** used to represent the **transmission** of **contagious diseases**.
- The **model** divides the population into three compartments: **S** for **susceptible individuals**, **I** for **infected individuals**, and **R** for **recovered individuals**.
- The model is defined by the following **differential equations** where β is the **transmission rate** and γ is the **recovery rate**:
 - $\frac{dS}{dt} = -\beta \cdot S \cdot I$
 - $\frac{dI}{dt} = \beta \cdot S \cdot I - \gamma \cdot I$
 - $\frac{dR}{dt} = \gamma \cdot I$



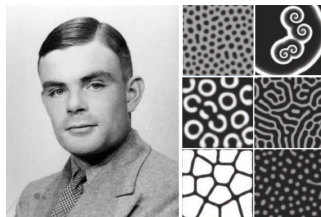
Chaotic Systems

- **Chaotic Systems** are a class of **dynamical systems** that exhibit **sensitive dependence** on initial conditions.
- This means that the **future behavior** of the system is **highly sensitive** to the **initial conditions**.
- The **Lorenz System** is a well-known example of a **chaotic system**.
- Using **cellular automata** to simulate **chaotic systems** is common practice. Many **fractals** can be created using **chaotic rules**.



Turing Morphogenesis

- **Turing Morphogenesis** is a theory of **biological development** that explains how **spatial patterns** form in **living organisms**.
- The theory is based on the idea that **chemical signals** can interact to create **spatial patterns** in living organisms.
- The **reaction-diffusion** model is commonly used to simulate **Turing morphogenesis**.
- The model is defined by a set of **reaction** and **diffusion** equations that describe how the **chemical signals** interact.



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What are Digital Twins?

- A **Digital Twin** is a *virtual model* that accurately reflects a physical object or **system**.
- **Digital twins** enable **real-time monitoring**, **simulation**, and **analysis** of the physical counterpart.
- They are widely used in **manufacturing**, **smart cities**, **healthcare**, and **energy systems**.
- By integrating sensor data, **digital twins** help in **predictive maintenance** and **process optimization**.



Applications of Digital Twins

- **Manufacturing:** Monitor equipment performance and **predict failures**.
- **Urban Planning:** Model city infrastructure to **optimize traffic flow** and resource allocation.
- **Healthcare:** **Simulate** patient-specific **treatments** and optimize medical procedures.
- **Energy Systems:** Analyze **grid performance** and integrate renewable energy sources.



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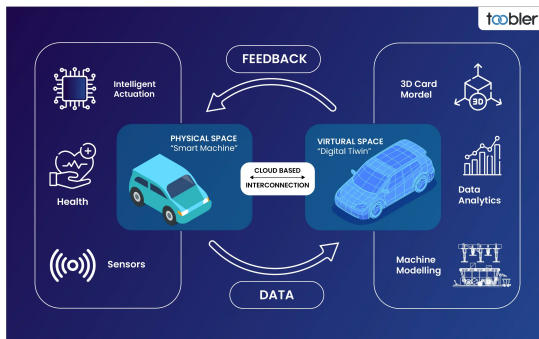
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Benefits of Digital Twins

- Enhanced **decision-making** through **real-time data analysis**.
- Improved **efficiency** and **reduced operational costs**.
- *Better understanding* of **system behavior** and **performance characteristics**.
- Enhanced **collaboration** between teams and stakeholders.



Challenges of Digital Twins

- Data **integration** and **management**.
- Ensuring data security and privacy.
- Maintaining the accuracy of the digital twin.
- Managing the complexity of the physical system.
- Ensuring interoperability between different systems and platforms.
- Scalability to accommodate large-scale systems.
- Cost of implementation and maintenance.
- User training and adoption.
- Regulatory compliance and standardization.
- Ethical considerations in data usage and analysis.
- Environmental impact of digital twin technologies.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

